

Networks Lab Assignment 1 Report

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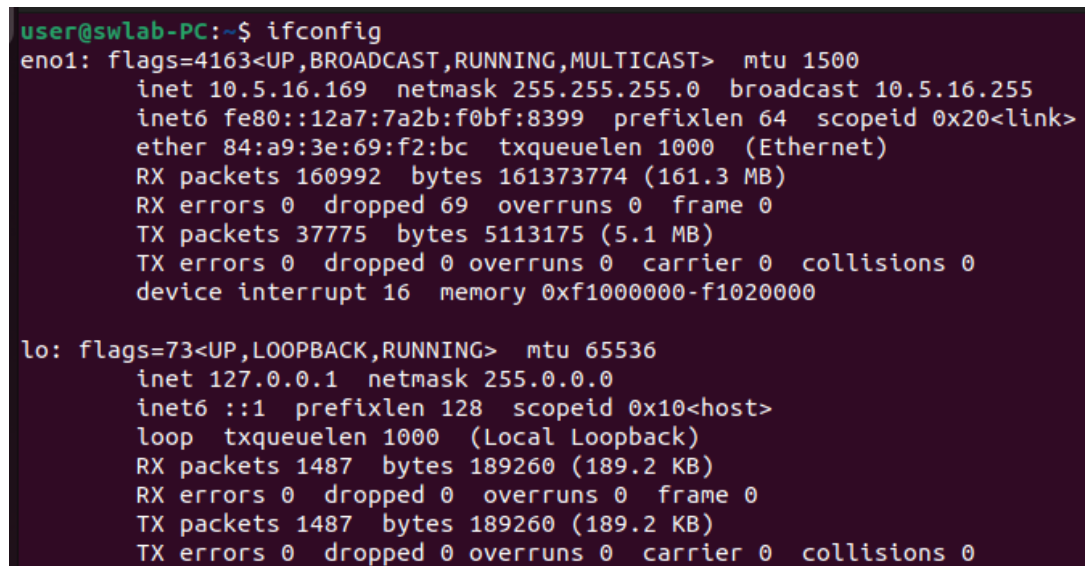
Part 1: Networking Tools

1 Task 1: Find the IP Address, Subnet Mask, and Network ID

Using the `ifconfig` command, we identified the following:

- **IP Address of the Machine:** 10.5.16.169
- **Subnet Mask:** 255.255.255.0
- **Network ID:** 10.5.16.0

Screenshot of the output:



```
user@swlab-PC:~$ ifconfig
eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.5.16.169 netmask 255.255.255.0 broadcast 10.5.16.255
    inet6 fe80::12a7:7a2b:f0bf:8399 prefixlen 64 scopeid 0x20<link>
    ether 84:a9:3e:69:f2:bc txqueuelen 1000 (Ethernet)
    RX packets 160992 bytes 161373774 (161.3 MB)
    RX errors 0 dropped 69 overruns 0 frame 0
    TX packets 37775 bytes 5113175 (5.1 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xf1000000-f1020000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1487 bytes 189260 (189.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1487 bytes 189260 (189.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 1: Output of `ifconfig` showing IP address, subnet mask, and network ID.

2 Task 2: Resolve Domain Names using nslookup

Using `nslookup`, the IP addresses for `www.google.com` and `www.facebook.com` were obtained:

- IP Address of **www.google.com**: 142.251.42.4
- IP Address of **www.facebook.com**: 31.13.79.35

When the DNS server address was changed to 172.16.1.164, 172.16.1.180, 172.16.1.165, and 172.16.1.166, the IP address of **www.google.com** was observed to change. This behavior is due to the load-balancing mechanisms used by DNS servers. These mechanisms ensure that traffic is distributed efficiently across multiple servers, improving performance, reliability, and redundancy.

```
dadi@RogStrix:~$ nslookup www.google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.251.42.4
Name:   www.google.com
Address: 2404:6800:4009:831::2004

dadi@RogStrix:~$ ^C
dadi@RogStrix:~$ nslookup www.facebook.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de
```

Figure 2: IP address associated with **www.google.com** and **www.facebook.com**

```
dadi@RogStrix:~$ nslookup www.google.com 172.16.1.164
Server:      172.16.1.164
Address:     172.16.1.164#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.250.67.228
Name:   www.google.com
Address: 2404:6800:4009:814::2004
```

Figure 3: IP address of **www.google.com** with DNS address 172.16.1.164

```
dadi@RogStrix:~$ nslookup www.google.com 172.16.1.180
Server:      172.16.1.180
Address:     172.16.1.180#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.250.192.132
Name:   www.google.com
Address: 2404:6800:4009:814::2004
```

Figure 4: IP address of **www.google.com** with DNS address 172.16.1.180

```
dadi@RogStrix:~$ nslookup www.google.com 172.16.1.165
Server:      172.16.1.165
Address:     172.16.1.165#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.251.42.4
Name:   www.google.com
Address: 2404:6800:4009:82f::2004
```

Figure 5: IP address of **www.google.com** with DNS address 172.16.1.165

```
dadi@RogStrix:~$ nslookup www.google.com 172.16.1.166
Server:      172.16.1.166
Address:     172.16.1.166#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.250.182.196
Name:   www.google.com
Address: 2404:6800:4009:81e::2004
```

Figure 6: IP address of **www.google.com** with DNS address 172.16.1.166

DNS	IP address
172.16.1.164	142.250.67.228
172.16.1.180	142.250.192.132
172.16.1.165	142.251.42.4
172.16.1.166	142.250.182.196

3 Task 3: Ping Command with Different Packet Sizes

Using the `ping` command, packets were sent to a friend's machine with varying sizes:

- Friend's IP Address: 10.5.16.152
- Packet Sizes: 64, 128, 512 bytes.
- Timeout: 100ms.

Results Table

The results of the `ping` command with varying packet sizes are summarized below:

Packet Size (bytes)	Packet Loss (%)	Round-Trip Time (min/avg/max/stddev)
64	0	0.948 / 1.311 / 1.636 / 0.182
128	0	1.025 / 1.261 / 1.513 / 0.161
512	0	1.203 / 1.404 / 1.674 / 0.122

Table 1: Summary of `ping` results for different packet sizes.

Screenshots

The following images show the `ping` command outputs for each packet size:

```
rtt min/avg/max/mdev = 0.669/1.209/1.857/0.222 ms
dadi@RogStrix: $ ping 10.5.16.152 -s 64 -W 100
PING 10.5.16.152 (10.5.16.152) 64(92) bytes of data:
72 bytes from 10.5.16.152: icmp_seq=1 ttl=64 time=1.19 ms
72 bytes from 10.5.16.152: icmp_seq=2 ttl=64 time=1.54 ms
72 bytes from 10.5.16.152: icmp_seq=3 ttl=64 time=1.09 ms
72 bytes from 10.5.16.152: icmp_seq=4 ttl=64 time=1.23 ms
72 bytes from 10.5.16.152: icmp_seq=5 ttl=64 time=1.28 ms
72 bytes from 10.5.16.152: icmp_seq=6 ttl=64 time=1.20 ms
72 bytes from 10.5.16.152: icmp_seq=7 ttl=64 time=1.03 ms
72 bytes from 10.5.16.152: icmp_seq=8 ttl=64 time=1.44 ms
72 bytes from 10.5.16.152: icmp_seq=9 ttl=64 time=1.46 ms
72 bytes from 10.5.16.152: icmp_seq=10 ttl=64 time=1.09 ms
72 bytes from 10.5.16.152: icmp_seq=11 ttl=64 time=1.47 ms
72 bytes from 10.5.16.152: icmp_seq=12 ttl=64 time=1.43 ms
72 bytes from 10.5.16.152: icmp_seq=13 ttl=64 time=1.47 ms
72 bytes from 10.5.16.152: icmp_seq=14 ttl=64 time=1.17 ms
72 bytes from 10.5.16.152: icmp_seq=15 ttl=64 time=0.948 ms
72 bytes from 10.5.16.152: icmp_seq=16 ttl=64 time=1.57 ms
72 bytes from 10.5.16.152: icmp_seq=17 ttl=64 time=1.31 ms
72 bytes from 10.5.16.152: icmp_seq=18 ttl=64 time=1.31 ms
72 bytes from 10.5.16.152: icmp_seq=19 ttl=64 time=1.44 ms
72 bytes from 10.5.16.152: icmp_seq=20 ttl=64 time=1.44 ms
72 bytes from 10.5.16.152: icmp_seq=21 ttl=64 time=1.64 ms
72 bytes from 10.5.16.152: icmp_seq=22 ttl=64 time=1.50 ms
72 bytes from 10.5.16.152: icmp_seq=23 ttl=64 time=1.32 ms
72 bytes from 10.5.16.152: icmp_seq=24 ttl=64 time=1.03 ms
72 bytes from 10.5.16.152: icmp_seq=25 ttl=64 time=1.17 ms
72 bytes from 10.5.16.152: icmp_seq=26 ttl=64 time=1.36 ms
^C
--- 10.5.16.152 ping statistics ---
26 packets transmitted, 26 received, 0% packet loss, time 25044ms
rtt min/avg/max/mdev = 0.948/1.311/1.636/0.182 ms
dadi@RogStrix: $
```

Figure 7: Ping output for 64-byte packets.

```

dadi@RogStrix: $ ping 10.5.16.152 -s 128 -W 100
PING 10.5.16.152 (10.5.16.152) 128(156) bytes of data.
136 bytes from 10.5.16.152: icmp_seq=1 ttl=64 time=1.08 ms
136 bytes from 10.5.16.152: icmp_seq=2 ttl=64 time=1.39 ms
136 bytes from 10.5.16.152: icmp_seq=3 ttl=64 time=1.27 ms
136 bytes from 10.5.16.152: icmp_seq=4 ttl=64 time=1.11 ms
136 bytes from 10.5.16.152: icmp_seq=5 ttl=64 time=1.31 ms
136 bytes from 10.5.16.152: icmp_seq=6 ttl=64 time=1.15 ms
136 bytes from 10.5.16.152: icmp_seq=7 ttl=64 time=1.20 ms
136 bytes from 10.5.16.152: icmp_seq=8 ttl=64 time=1.51 ms
136 bytes from 10.5.16.152: icmp_seq=9 ttl=64 time=1.51 ms
136 bytes from 10.5.16.152: icmp_seq=10 ttl=64 time=1.03 ms
136 bytes from 10.5.16.152: icmp_seq=11 ttl=64 time=1.43 ms
136 bytes from 10.5.16.152: icmp_seq=12 ttl=64 time=1.40 ms
136 bytes from 10.5.16.152: icmp_seq=13 ttl=64 time=1.25 ms
136 bytes from 10.5.16.152: icmp_seq=14 ttl=64 time=1.04 ms
^C
--- 10.5.16.152 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13022ms
rtt min/avg/max/mdev = 1.025/1.261/1.513/0.161 ms
dadi@RogStrix: $

```

Figure 8: Ping output for 128-byte packets.

```

dadi@RogStrix: $ ping 10.5.16.152 -s 512 -W 100
PING 10.5.16.152 (10.5.16.152) 512(540) bytes of data.
520 bytes from 10.5.16.152: icmp_seq=1 ttl=64 time=1.29 ms
520 bytes from 10.5.16.152: icmp_seq=2 ttl=64 time=1.28 ms
520 bytes from 10.5.16.152: icmp_seq=3 ttl=64 time=1.67 ms
520 bytes from 10.5.16.152: icmp_seq=4 ttl=64 time=1.25 ms
520 bytes from 10.5.16.152: icmp_seq=5 ttl=64 time=1.45 ms
520 bytes from 10.5.16.152: icmp_seq=6 ttl=64 time=1.53 ms
520 bytes from 10.5.16.152: icmp_seq=7 ttl=64 time=1.26 ms
520 bytes from 10.5.16.152: icmp_seq=8 ttl=64 time=1.48 ms
520 bytes from 10.5.16.152: icmp_seq=9 ttl=64 time=1.38 ms
520 bytes from 10.5.16.152: icmp_seq=10 ttl=64 time=1.35 ms
520 bytes from 10.5.16.152: icmp_seq=11 ttl=64 time=1.59 ms
520 bytes from 10.5.16.152: icmp_seq=12 ttl=64 time=1.46 ms
520 bytes from 10.5.16.152: icmp_seq=13 ttl=64 time=1.31 ms
520 bytes from 10.5.16.152: icmp_seq=14 ttl=64 time=1.33 ms
520 bytes from 10.5.16.152: icmp_seq=15 ttl=64 time=1.56 ms
520 bytes from 10.5.16.152: icmp_seq=16 ttl=64 time=1.34 ms
520 bytes from 10.5.16.152: icmp_seq=17 ttl=64 time=1.52 ms
520 bytes from 10.5.16.152: icmp_seq=18 ttl=64 time=1.39 ms
^C
--- 10.5.16.152 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 17033ms
rtt min/avg/max/mdev = 1.203/1.404/1.674/0.122 ms
dadi@RogStrix: $

```

Figure 9: Ping output for 512-byte packets.

4 Task 4: Traceroute Command

Using the `traceroute` command, the path to `www.google.com` was analyzed. The number of hosts involved in the path from source to destination was counted.

Summary of Results:

- Number of Hosts: 15
- Observed “* * *” in intermediate hops: These indicate timeouts or unreachable hosts, often caused by network policies or firewalls or TTL expiration without response

Screenshot of the output:

```

dadi@RogStrix: $ traceroute www.google.com
traceroute to www.google.com (142.250.194.196), 30 hops max, 60 byte packets
 1 _gateway (10.5.16.2) 0.978 ms 0.919 ms 0.882 ms
 2 10.120.2.33 (10.120.2.33) 0.838 ms 0.802 ms 0.765 ms
 3 10.255.1.3 (10.255.1.3) 3.379 ms 2.751 ms 2.701 ms
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 72.14.204.62 (72.14.204.62) 37.283 ms 37.636 ms 142.250.172.80 (142.250.172.80) 40.549 ms
 9 * * *
10 209.85.142.84 (209.85.142.84) 36.026 ms 142.250.235.10 (142.250.235.10) 41.769 ms 142.251.69.104 (142.251.69.104) 44.855 ms
11 192.178.110.248 (192.178.110.248) 41.733 ms 142.250.226.134 (142.250.226.134) 43.374 ms 192.178.110.244 (192.178.110.244) 43.341 ms
12 142.251.48.137 (142.251.48.137) 61.637 ms 192.178.251.217 (192.178.251.217) 56.788 ms 172.253.68.120 (172.253.68.120) 38.351 ms
13 192.178.251.217 (192.178.251.217) 57.690 ms 172.253.51.137 (172.253.51.137) 62.233 ms 192.178.83.215 (192.178.83.215) 61.249 ms
14 142.251.52.207 (142.251.52.207) 62.164 ms 142.251.52.209 (142.251.52.209) 65.030 ms 209.85.250.57 (209.85.250.57) 67.653 ms
15 del12s07-in-f4.1e100.net (142.250.194.196) 59.114 ms 142.251.52.209 (142.251.52.209) 59.040 ms 59.591 ms
dadi@RogStrix: $

```

Figure 10: Output of traceroute for `www.google.com`.

Part 2: Packet Analysis

1. DNS Packets Analysis

- DNS Query/Response Protocol: UDP
- Source IP Address of DNS Query: 10.5.16.152
- Destination IP Address of DNS Query: 172.16.1.180
- Number of DNS Queries Sent: 1, packet no: 270
- DNS Server Replying with IP Address: 172.16.1.180

- Number of DNS Servers Involved: 1 (Not all DNS servers responded)
- Resource Records:

```

271.17.59 > Ethernet II, Src: HewlettP.69:f9:64 (84:a9:3e:69:f9:64), Dst: 98:88:55:9c:c4:4a (98:88:55:9c:c4:4a)
271.17.59 > Internet Protocol Version 4, Src: 192.168.1.100, Dst: 172.16.1.100
459.19.68 > User Datagram Protocol, Src Port: 42921, Dst Port: 53
451.19.68 > Domain Name System (query)
490.19.68 > Transaction ID: 0x7c0c
612.19.78 > Flags: Standard query
613.19.71 > Questions: 1
698.19.92 > Answer RRs: 0
705.19.93 > Authority RRs: 0
717.19.94 > Additional RRs: 1
718.19.94 > Queries
1369.19.47 > www.litkgp.ac.in: type A, class IN
1369.19.47 > Name: www.litkgp.ac.in
1404.19.78 > [Name Length: 16]
1405.19.77 > [Label Count: 4]
1428.19.88 > Type: A (Host Address) (1)
1429.19.88 > Class: IN (0x0001)
1448.20.08 > Additional records
1629.20.41 > <Root>: type OPT

```

Figure 11: DNS Query

```

270.17.943975401 19.5.18 > Additional RRs: 1
613.19.69313244 172.16.1 > Queries
459.19.68291594 19.5.18 > www.litkgp.ac.in: type A, class IN
451.19.68299815 19.5.18 > Name: www.litkgp.ac.in
490.19.686117639 172.16.1 > [Name Length: 16]
612.19.78929242 19.5.18 > [Label Count: 4]
613.19.71460226 172.16.1 > Type: A (Host Address) (1)
698.19.927464366 19.5.18 > Class: IN (0x0001)
705.19.93851221 172.16.1 > Answers
717.19.94951159 19.5.18 > www.litkgp.ac.in: type A, class IN, addr 172.16.3.10
718.19.94529656 172.16.1 > Name: www.litkgp.ac.in
1369.19.47531670 19.5.18 > Type: A (Host Address) (1)
1369.19.47531670 172.16.1 > Class: IN (0x0001)
1404.19.767487171 19.5.18 > Time to Live: 86400 (1 day)
1405.19.770976655 172.16.1 > Data Length: 4
1428.19.881924806 19.5.18 > Address: 172.16.3.10
1429.19.884497862 172.16.1 > Additional records
1448.20.080842381 172.16.1 > <Root>: type OPT
1629.20.414824541 19.5.18 > Name: <Root>

```

Figure 12: DNS Response

```

271.17.943975401 19.5.18 > 19.5.18.100 19.5.18.100 0 0 Standard query response 0x7c0c A www.litkgp.ac.in A 172.16.3.10 OPT
459.19.68291594 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
451.19.68299815 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
490.19.686117639 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
612.19.78929242 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
613.19.71460226 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
698.19.927464366 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
705.19.93851221 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
717.19.94951159 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
718.19.94529656 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1369.19.47531670 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1369.19.47531670 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1404.19.767487171 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1405.19.770976655 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1428.19.881924806 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1429.19.884497862 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1448.20.080842381 172.16.1.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT
1629.20.414824541 19.5.18.100 172.16.1.100 0 0 Standard query 0x7c0c A www.litkgp.ac.in OPT

```

Figure 13: nslookup response

2. Web Traffic (HTTP) Analysis

- Number of HTTP Packets Exchanged: 12
- HTTP Request: GET /~grovesd/ HTTP/1.1
- HTTP Response: HTTP/1.1 200 OK (text.html)

```

Wireshark - Packet 5049 - eno1
Frame 5049: 587 bytes on wire (4696 bits), 587 bytes captured (4696 bits) on interface eno1, id 0
Ethernet II, Src: HewlettP.69:f9:64 (84:a9:3e:69:f9:64), Dst: 98:88:55:9c:c4:4a (98:88:55:9c:c4:4a)
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 172.16.1.100
Transmission Control Protocol, Src Port: 30774, Dst Port: 80, Seq: 1, Ack: 1, Len: 441
Hypertext Transfer Protocol
GET /~grovesd/ HTTP/1.1
[Expert Info (Chat/Sequence): GET /~grovesd/ HTTP/1.1]
Request Method: GET
Request URI: /~grovesd/
Request Version: HTTP/1.1
Host: web.simmons.edu
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/84.0.0.0 Safari/537.3
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en,en-US;q=0.9,en;q=0.8
[Full request URI: http://web.simmons.edu/~grovesd/]
HTTP request 1/4
[Request in frame 5056]
Next request in frame 5056

```

(a) HTTP Request

```

Wireshark - Packet 5056 - eno1
Frame 5056: 2321 bytes on wire (18568 bits), 2321 bytes captured (18568 bits) on interface eno1, id 0
Ethernet II, Src: HewlettP.69:f9:64 (84:a9:3e:69:f9:64), Dst: HewlettP.69:f9:64 (84:a9:3e:69:f9:64)
Internet Protocol Version 4, Src: 69.43.111.82, Dst: 192.168.1.100
Transmission Control Protocol, Src Port: 80, Dst Port: 30774, Seq: 1, Ack: 442, Len: 2255
Hypertext Transfer Protocol
HTTP/1.1 200 OK
[Expert Info (Chat/Sequence): HTTP/1.1 200 OK]
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
Date: Thu, 16 Jan 2025 17:01:04 GMT
Server: Apache/2.4.18
Last-Modified: Tue, 03 Sep 2019 00:33:59 GMT
ETag: "7b2-5918b3d6d6b0c"
Accept-Ranges: bytes
Content-Length: 1970
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
[HTTP response 1/4]
[Time since request: 0.232135865 seconds]
[Request in frame 5049]
Next request in frame 5049
Next request in frame 5056
[Request URI: http://web.simmons.edu/favicon.ico]
File Data: 1970 bytes
Line-based text data: text/html (57 lines)

```

(b) HTTP Response

```

File Edit View Go Capture Analysis Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
1 454.92 192.168.1.100 192.168.1.100 HTTP 154 GET / HTTP/1.1
2 459.20 192.168.1.100 192.168.1.100 HTTP 154 GET / HTTP/1.1
3 504.92 69.43.111.82 192.168.1.100 HTTP 2321 GET /~grovesd/ HTTP/1.1
4 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
5 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
6 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
7 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
8 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
9 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
10 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
11 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
12 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
13 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
14 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
15 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
16 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
17 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
18 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
19 505.60 69.43.111.82 192.168.1.100 HTTP 154 GET / HTTP/1.1
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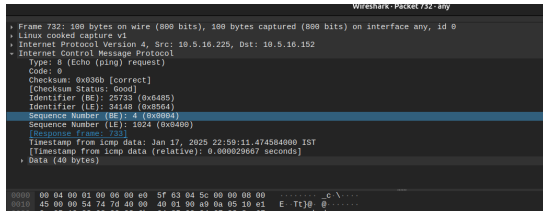
```

(c) HTTP Request 3.

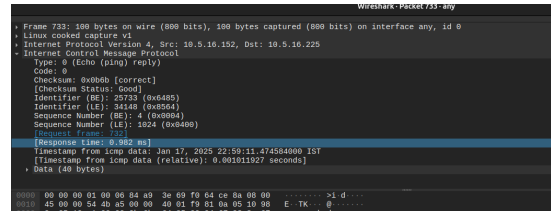
Figure 14: HTTP Request and Response for <http://web.simmons.edu/~grovesd/>.

3. ICMP Traffic (Ping/Traceroute)

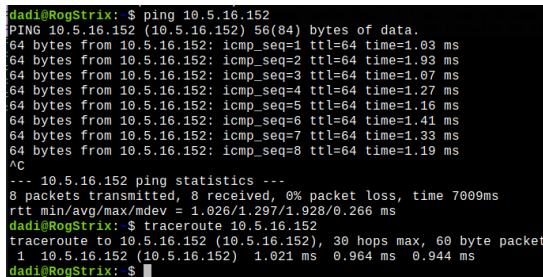
- Ping to Friend's IP: 10.5.16.152
- Traceroute to Friend's IP: 10.5.16.152
- Ping to Unreachable Host: 192.168.1.100



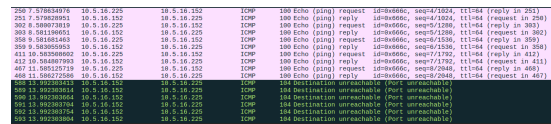
(a) request to 10.5.16.152



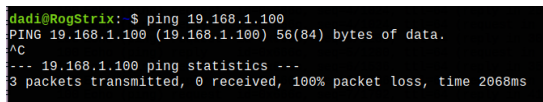
(b) response packet from 10.5.16.152



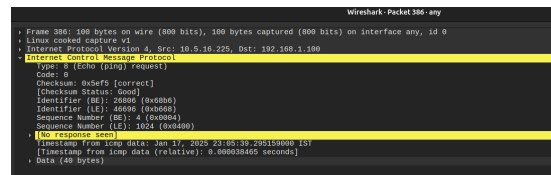
(c) ping and traceroute to 10.5.16.152



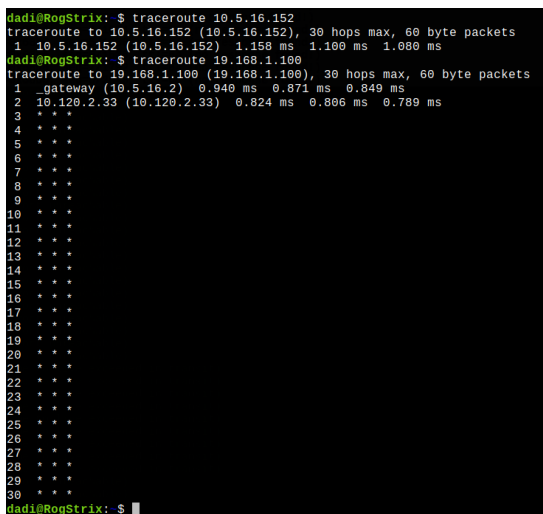
(d) "ping" and "traceroute" captured in Wireshark.



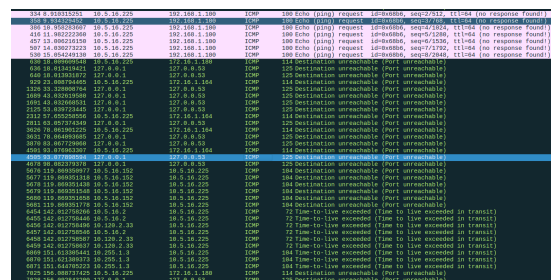
(e) ping to unreachable host(19.168.1.100)



(f) packet to unreachable host - no response



(g) traceroute to 10.5.16.152(reachable host) and 19.168.1.100(unreachable host)



(h) traceroute to reachable and unreachable hosts

Figure 15: ICMP Packets for Ping/Traceroute.

For the first traceroute to the reachable host `10.5.16.152`, the packets were successfully routed to the destination in the first hop. This indicates that the destination IP is directly accessible on the local network. The ICMP Echo Request packets sent from the host were followed by the ICMP Echo Reply packets from the destination with minimal latency, confirming a direct and fast connection.

For the second traceroute to the unreachable host `19.168.1.100`, the process initially passed through the local gateway (`10.5.16.2`) and the first two hops responded successfully. However, after the second hop, all subsequent hops showed no response, marked by `* * *`. This suggests that the destination host is unreachable, either due to non-existence in the network, a firewall blocking ICMP replies, or a routing issue preventing the packets from reaching the destination. In Wireshark, while the ICMP Echo Request packets were sent, no ICMP Time Exceeded or Echo Reply packets were received beyond the second hop. This absence of responses indicates that the packets did not proceed beyond the first few hops, likely due to network or firewall filtering mechanisms.